

Alexandre Andre

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I am a Ph.D. student in CIS at the University of Pennsylvania, advised by Prof. Eva L. Dyer, working at the intersection of deep learning and neuroscience. My research centers on representation learning for invasive neural recordings, spanning benchmarking, large-scale pretraining, and interpretability, with an emphasis on models that generalize across subjects, sessions, brain regions, and recording technologies. I also apply these methods to other biosignals for clinical use.

EDUCATION

University of Pennsylvania

Ph.D. in Computer and Information Science · Advisor: Prof. Eva L. Dyer

Aug 2025 – Present

Philadelphia, PA, USA

Georgia Institute of Technology

M.S. in Computer Science · Machine Learning specialization · GPA 4.0/4.0

Jan 2024 – May 2025

Atlanta, GA, USA

University of Technology of Compiègne (UTC)

M.Eng. in Computer Science · GPA 5.0/5.0

Sep 2019 – Apr 2025

Compiègne, France

Politecnico di Milano

Visiting Student, Computer Systems Engineering Department

Feb – Jul 2021

Milan, Italy

PUBLICATIONS

* equal contribution † co-supervision

[1] BrainWideBench: Benchmarking Large-Scale Pretraining and Across-Animal Transfer in Multi-Region Neural Recordings

Andre, A.* , Mahato, S.P.* , Arora, V., Balaji, K., Lachi, D., Krishna, N.H., Xiao, J., Zhang, Y., Mao, X., Ma, W., Yu, H., The International Brain Laboratory (IBL), Birman, D., Bonacchi, N., Chapuis, G.A., Catarino, J.A., Davatolhagh, F., Faulkner, M., Freitas-Silva, L., Hu, F., Huntenburg, J.M., Khanal, A., Laranjeira, I., Lau, P., Meijer, G.T., Miska, N.J., Noel, J.-P., Pan-Vazquez, A., Raiser, G., Rossant, C., Socha, K.Z., Urai, A.E., Wells, M.J., West, S.J., Winter, O., Richards, B., Lajoie, G., Hurwitz, C., Azabou, M., Whiteway, M.R.† , Paninski, L.† , Dyer, E.L.†
Under review, Conference on Neural Information Processing Systems (NeurIPS), 2026.

[2] Projection-specific Routing of Odor Information in the Olfactory Cortex

Daste, S.* , Pham, T.H.* , Seppo, M., Andre, A., Srinivasan, S., Xiao, J., Sattin, A., Nardin, C., Fellin, T., Franks, K.M., Dyer, E.L., Fleischmann, A.
Current Biology, 2026. In press

[3] PRISM: A Hierarchical Multiscale Approach for Time Series Forecasting

Chen, Z., Andre, A., Ma, W., Knight, I.J., Shuvaev, S.A., Dyer, E.L.
Preprint, 2026.

[4] Revealing Potential Biases in LLM-Based Recommender Systems in the Cold Start Setting

Andre, A.* , Roy, G.* , Dyer, E.L., Wang, K.
2nd RecSys Workshop on Evaluating and Applying Recommendation Systems with Large Language Models (EARL), 2025. Oral

[5] A Scalable Self-Supervised Method for Modeling Human Intracranial Recordings During Natural Behavior

Mahato, S.P.* , Xiao, J.* , Andre, A., Chau, G., Ma, W., Knight, I.J., Nguyen, D., Hu, L.J., Brunton, B.W., Beauchamp, M.S., Pesaran, B., Shuvaev, S.A., Dyer, E.L.
NeurIPS Workshop on Foundation Models for the Brain and Body (BrainBodyFM), 2025. Oral

[6] Neural Encoding and Decoding at Scale

Zhang, Y.* , Wang, Y.* , Azabou, M., Andre, A., Wang, Z., Lyu, H., IBL, Dyer, E.L., Paninski, L., Hurwitz, C.
International Conference on Machine Learning (ICML), 2025. Spotlight

RESEARCH EXPERIENCE

NerDS Lab

Oct 2024 – Present

Graduate Research Assistant, supervised by Prof. Eva L. Dyer

Atlanta, GA & Philadelphia, PA, USA

- **Representation learning for invasive neural recordings across species and recording modalities.**
- Led the design of [BrainWideBench](#), a benchmark for neuro foundation models built on the IBL Brainwide Map (139 mice, 276 brain regions, 452 sessions, 830K+ unit-hours), with three task suites: behavior decoding, neural dynamics prediction, and anatomical classification. Evaluated 6 large-scale pretrained models and tuned 18 single-session baselines, removing the need to rerun baselines and preprocessing when developing new methods.
- Collaborated on deep learning methods across invasive neural modalities, from large-scale pretraining (multimodal mouse spikes and behavior, [ICML 2025](#); human iEEG, [NeurIPS Workshop 2025](#)) to interpretability (calcium imaging, [Current Biology 2026](#)).

Heudiasyc Lab, UMR-CNRS 7253

Sep – Dec 2023

Graduate Research Assistant, supervised by Prof. Julien Moreau and Prof. Philippe Xu

Compiègne, France

- **Localization prediction in a weakly supervised setting.**
- Adapted [DETR](#) to detect key points, specifically the ground contact points of poles, in complex urban environments.

INDUSTRY EXPERIENCE

Deezer

Jun – Nov 2024

Machine Learning Engineer Intern, supervised by [Théo Bontempelli](#)

Paris, France

- **Modeling user musical tastes with transformers in a large-scale environment.**
- Built transformer-based user embeddings by aggregating song embeddings from listening histories.
- Improved accuracy@1K by 15% relative to the production baseline on a catalog of tens of millions of tracks and users; handed off the model for a production A/B test.

Wise Systems

Sep 2022 – Feb 2023

Cloud/DevOps Engineer Intern

Montreal, Canada

- **CI/CD improvements to Infrastructure-as-Code deployed with Terraform.**
- Streamlined infrastructure management by deploying [Atlantis](#), a Terraform pull request automation tool.

OPEN-SOURCE SOFTWARE

`torch_brain`

2024 – Present

Co-lead developer and technical committee member

[torchbrain.org](#)

- **A PyTorch library for building efficient training pipelines and models on neural data (90+ GitHub stars).**
- Guide the technical roadmap and standards for reusable data and model interfaces across heterogeneous neurophysiology datasets, enabling scalable training on large neural recordings.
- Review and integrate community contributions; coordinate testing, documentation, and software releases.
- Drive outreach and adoption through tutorials, talks, and community engagement across neuroscience and ML labs.

TALKS

NeuroDataReHack

Jun 2026

Building Neuro Foundation Models with `torch_brain`

HHMI Janelia Research Campus, VA, USA

ARNI Frontier Models for Neuroscience and Behavior Working Group

Feb 2026

[BrainWideBench](#): Benchmarking large-scale pretraining on multi-region neural recordings

New York, NY, USA

EARL Workshop, RecSys

Sep 2025

Revealing Potential Biases in LLM-Based Recommender Systems in the Cold Start Setting

Prague, Czech Republic

ACADEMIC SERVICE

Co-organizer , NeurIPS Workshop on Foundation Models for the Brain and Body	2026
Co-organizer , ARNI Frontier Models for Neuroscience and Behavior Working Group	2026
Reviewer , Conference on Neural Information Processing Systems (NeurIPS)	2026
Reviewer , Computational and Systems Neuroscience (COSYNE)	2026
Reviewer , NeurIPS Workshop on Foundation Models for the Brain and Body (BrainBodyFM)	2025

MENTORSHIP

Max Mercado , M.S. in Computer and Information Science, University of Pennsylvania	Fall 2026
Reid Laughton , M.S. in Computer and Information Science, University of Pennsylvania	Summer 2026

TEACHING EXPERIENCE

Teaching Assistant , CIS 6250: Theory of Machine Learning, University of Pennsylvania	Fall 2025
Tutor , undergraduate statistics, probability, analysis, and linear algebra, UTC	Fall 2023

TECHNICAL SKILLS AND LANGUAGES

Programming	Python (PyTorch, scikit-learn), W TeX, Go, C++, SQL, C, R, MATLAB
Technologies	AWS, GCP, Git, GitHub, SLURM/HPC, W&B, GitLab CI/CD, Docker, Terraform
Languages	English (full working proficiency), French (native), Italian (intermediate)